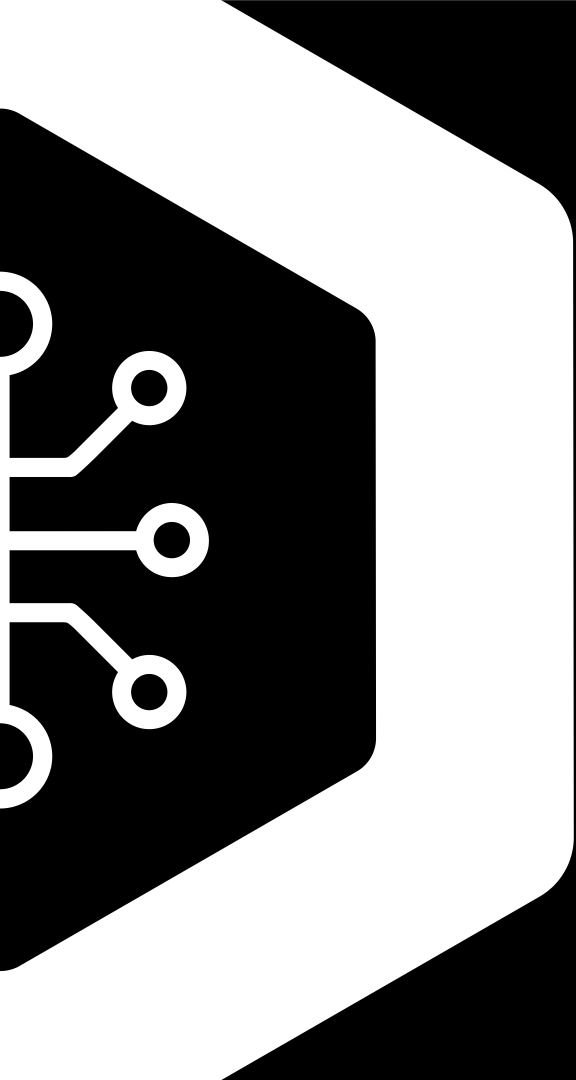




Open Silicon in Action: What's New from CHIPS Alliance

DAC Open-Source EDA BoF, Long Beach, 2026-07-28

Michael Gielda, mgielda@antmicro.com



CHIPS Alliance

Part of the Linux Foundation, CA works on all the necessary ingredients of open source silicon designs:

- Open source IP (CPUs, I/O, security etc.)
- Open source tooling (design, DV, productivity, simulation etc.)
- Open source PDKs (re-launching soon!)

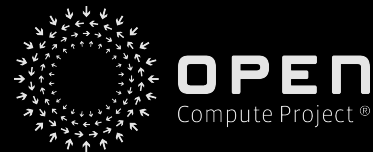
CHIPS Alliance brings together industrial adopters and tool implementers across the entire stack.



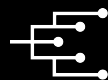
CHIPS supports a broad collaboration

CHIPS is actively collaborating with other open source organizations in the open silicon space

- OCP hosts the Caliptra and L.O.C.K. specifications, and CHIPS hosts implementations
- CHIPS supports FOSSi Foundation's conferences and uses / contributes to its software projects like cocotb
- OpenHW group uses CHIPS' work such as RISC-V DV and Verilator improvements to help with verification workflows
- key CHIPS persons participate in the OpenROAD Initiative which complements CHIPS' open tooling efforts



CHIPS Alliance Projects (chipsalliance.org/projects)

 Caliptra

CHISEL

 F4PGA

 VeeR

 SV Tools
PROJECT

OpenPRoT

Qualcomm joins CHIPS ALLIANCE

Qualcomm [joined as Platinum Member](#) in July

- Brings decades of experience designing premium-tier SoCs for mobile, automotive, and data center applications
- Lu Dai, Vice President of Technical Standards at Qualcomm joins the Governing Board
 - Deep expertise in design verification, SoC methodologies, and the complex coordination required between design houses and foundries
 - Chairman of the Board for both Accelera and RISC-V



The Qualcomm logo, consisting of the word "Qualcomm" in a white, sans-serif font, centered on a dark background.

SV Tools Project updates

Now officially [re-formed as a CA Project](#)

- Suite of open source tools supporting SystemVerilog and the most common SV design verification methodology, UVM
- sv-tests constantly evolving, became a staple benchmark for open source tools
- Antmicro adding features needed for RISC-V-DV support into Verilator (old version used by OHG already supported)



SV-TESTS

VERIBLE

SYNLIG

RISC-V-DV

SV Tools Project updates

Growing engagement from industry and academia

- Bi-weekly TSC meetings with attendees across the open source tooling landscape: slang, Verilator, Surfer, CIRCT, coverage tooling...
- Visit <https://github.com/chipsalliance/sv-tools>
- Talk by Krzysztof Bieganski (Antmicro) later today!



...

VeeR with DCLS

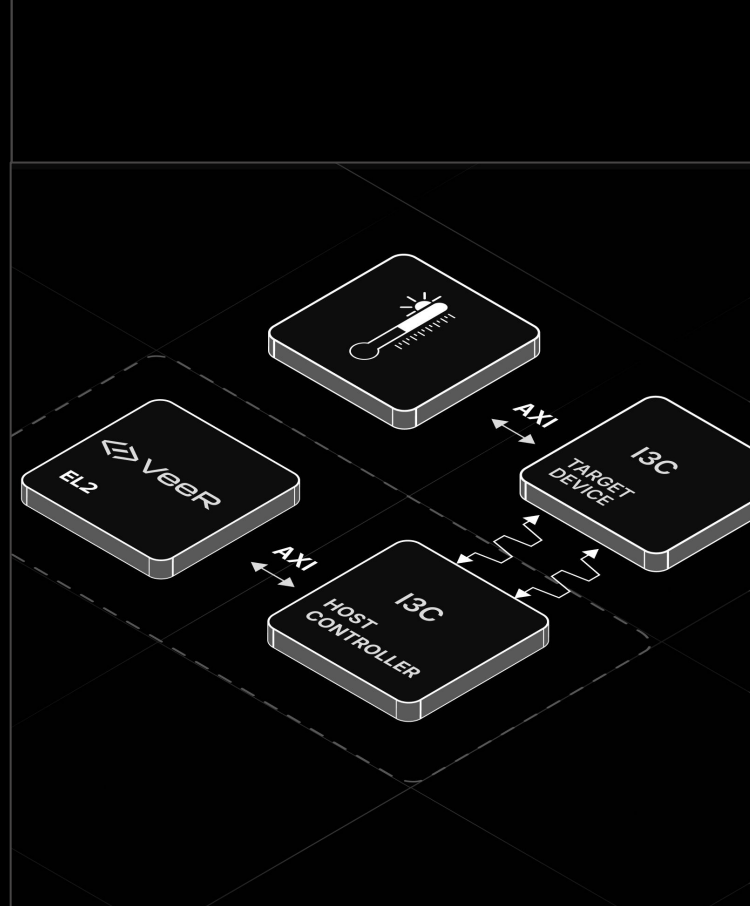
- VeeR: family of RISC-V-based cores
 - EL2 32-bit variant used in Caliptra RoT
 - [Guineveer](#) provides a reference SoC
- DCLS support merged into main:
<https://github.com/chipsalliance/Cores-VeeR-EL2/pull/354>
- Upcoming: TMR!
- [Read more](#) on the CA Blog
- Visit <https://github.com/chipsalliance/VeeR>
- Bi-weekly TSC meetings with Antmicro, Google, HM, lowRISC, and growing



I3C Host Controller

CHIPS I3C core gets host controller, as well as:

- Full device DV coverage
- Hardware testing with AMD ZynqMP ZCU106 evaluation board
 - Host controller connected to Antmicro's PMOD I3C Sensor Board
 - Demonstrated full 12.5MHz communication speeds against off-the-shelf silicon
- Visit <https://github.com/chipsalliance/i3c-core>
- [Read more](#) on Antmicro's Blog



Chisel

7 releases this year

- Clock Domains - synchronous/rational clock domain modeling with ClockDomain class
- Ongoing effort to switch the project to Scala 3
- Toolchain updates
- New language features - **past** operator, **asReset** cast, string concatenation, **DynamicGroup** for runtime option groups, **noModulePrefix**/custom extmodule **defname**

The logo for CHISEL, featuring the word in a stylized, white, blocky font with small circles at the ends of the letters, set against a dark background.

Caliptra RoT

- OCP L.O.C.K. (Layered Open-source Cryptographic Key management), Caliptra 2.1+, provides secure key management for Data-At-Rest protection in self-encrypting storage devices
- ROM SDK 2.1.0 and 2.0.1 releases
 - [See changelogs](#)
- Visit <https://github.com/chipsalliance/caliptra>



OpenPRoT

An open, standardized firmware stack for all platform security operations

- Onboarded as a CA project in Q4 2025
- Core architecture and portability layers driven by AMD and Google
- Adheres to industry-standard specifications from OCP, DMTF, and PCI-SIG
- Complements CA's Caliptra RoT
- Visit <https://github.com/OpenPRoT>

OpenPRoT

2026 events

Sponsoring , exhibiting and presenting at industry events:

- FOSSi Foundation's annual conferences
 - Latch-Up, May 1-3, Ontario, Canada
 - **Upcoming:** ORConf, September 11-13, Ghent, Belgium
- OCP EMEA, April 29-30, Barcelona, Spain
- RISC-V Summit Europe, June 8-11, Bologna, Italy
- DAC (sponsoring the BoF again this year!)
- **Upcoming:** OCP Global, October 12-15, San Jose, CA



Latch-Up



RISC-V
SUMMIT



ORConf 2026

How to get involved



MAILING LISTS

Our primary channel for announcements & technical discussion:
lists.chipsalliance.org



SLACK

Chat with the community:
slack-invite.chipsalliance.org



LINKEDIN

Connect for news & updates:
[@chipsalliance](https://www.linkedin.com/company/chipsalliance)



COMMUNITY CALENDAR

Meeting schedules, open to all:
calendar.chipsalliance.org



GITHUB

Browse code, open issues, send pull requests:
[@chipsalliance](https://github.com/chipsalliance)



BECOME A MEMBER

Join as an organization:
chipsalliance.org/join



**THANK YOU
FOR YOUR ATTENTION!**

